

Global Student Digital Behavior & Brain Rot Index Analysis

Insights Report — Questions & Findings

A structured, SQL-driven exploration of how digital habits — social media, short-form video, and online learning — relate to academic outcomes, wellbeing, and financial risk among students worldwide.

PREPARED BY

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About This Report

This report documents the exploratory data analysis performed on the student_behavior table (database: student_brainrot), a global survey of student digital habits. All cleaning, transformation, and analysis were performed natively in SQL (MySQL), and are organized below into fifteen questions, each presented with its finding.

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ATTRIBUTES PER STUDENT RECORD	ANALYSIS QUESTIONS	DUPLICATE RECORDS FOUND

Data Preparation Summary

- Checked student_id for duplicates across the full table — none were found.
- Audited every column for NULL or blank values using a dynamically generated SQL statement built from information_schema.columns.
- Found 6,262 blank entries in brain_rot_level; rebuilt the field from brain_rot_index using threshold logic (Low ≤ 19, Medium ≤ 39, High ≤ 59).
- Engineered a new digital_addiction_level field (Low / Moderate / High) from digital_addiction_score bands to support productivity analysis.

THEME 01

Demographics & Usage Patterns

Who the students are, and how their basic usage varies

Q1

What is the overall distribution of students by country, gender, age, and development level?

FINDING

The dataset spans a wide range of countries, age groups, genders, and development levels, giving the sample broad global coverage. This balanced spread across demographic categories provides a reliable backbone for comparing digital behavior across different population segments later in the report.

Q2

Which countries have the highest number of students in the dataset?

FINDING

Cambodia has the highest total student count of any country in the dataset, identified by ranking country-level counts in descending order.

Q3

How does device usage differ across students based on their demographic characteristics?

FINDING

Students in urban areas show the highest device usage overall, ahead of their rural peers, measured by comparing student counts across the urban/rural segment.

Q4

How much time do students spend on social media, online learning, entertainment, and short-form video content?

FINDING

Social media and short-form video account for the largest share of students' daily screen time, followed by entertainment content, while online learning hours remain comparatively low. This points to entertainment-driven browsing, rather than structured study, as the dominant use of students' online time.

Q5

Which digital activity consumes the most average time per student?

FINDING

Social media is the single largest consumer of students' digital time on average, ahead of short-form video, entertainment content, online learning, and education content.

Q6

How does digital behavior vary across different income levels?

FINDING

Screen time varies by household income: middle-income students spend the most time on digital activity, low-income students come second, and high-income students spend the least.

THEME 02

Brain Rot Index & Digital Behavior

How usage intensity connects to the Brain Rot classification

Q7

What proportion of students fall into each brain rot level?

FINDING

The largest share of students falls into the Medium brain rot band, with smaller proportions classified as Low and High. This distribution suggests that moderate digital-behavior risk is the norm across the surveyed population, rather than an extreme concentrated at either end.

Q8

How does social media usage relate to the brain rot index?

FINDING

There is a clear positive relationship between the two: as social media hours increase, the average Brain Rot Index also increases, confirmed by grouping rounded social media hours and tracking the average index at each level.

Q9

Is short-form video consumption associated with higher brain rot levels?

FINDING

Yes. Average short-video hours rise steadily from the Low to the High brain rot band, mirroring the relationship already confirmed for social media, and reinforcing short-form video as a key driver of elevated brain rot scores.

Q10

How prevalent is late-night digital usage across students?

FINDING

A substantial share of students report frequent late-night device usage. This pattern tracks closely with higher Brain Rot Index scores and reduced sleep hours, making late-night usage one of the more actionable behavioral signals in the dataset.

Academic & Wellbeing Impact

Connecting digital habits to motivation and productivity

Q11

How does academic motivation vary across brain rot levels?

FINDING

Academic motivation is slightly higher among students with a Low Brain Rot level compared to those in the Medium or High bands.

Q12

Is higher digital addiction associated with lower productivity?

FINDING

Yes. Students with High digital addiction have the lowest average productivity score, while students in the Low addiction band score the highest. A new digital_addiction_level field (Low / Moderate / High, derived from score thresholds) was created specifically to support this comparison.

Q13

Which are the top 3 countries by Brain Rot Index within each development level?

FINDING

The top-ranked countries differ from one development tier to the next, rather than the same countries repeating across tiers. This indicates that elevated digital-behavior risk is driven by localized, country-specific factors rather than a uniform pattern tied only to a country's overall development status.

THEME 04

Spending & Risk Exposure

Financial behavior and safety-related exposure

Q14

How does digital spending vary across income levels?

FINDING

Digital spending scales with household income: high-income students spend the most, middle-income students spend a moderate amount, and low-income students spend the least — the reverse ordering from the screen-time pattern found in Q6.

Q15

Is exposure to adult content high or low among students?

FINDING

Adult content exposure across the student population is very low overall, based on counts of the `adult_content_exposure` flag, suggesting this is not currently a widespread risk factor within the surveyed group.

What the Findings Suggest

- Screen time and Brain Rot Index rise together with social media and short-form video use — a direct, monitorable signal.
- Academic motivation and productivity both decline as digital-behavior risk and addiction levels increase.
- Income shapes digital behavior in opposite directions: middle-income students spend more time online, while high-income students spend more money.

Suggested Next Steps

- Prioritize screen-time and sleep-hygiene guidance for High Brain Rot / High addiction students.
- Investigate country-level drivers behind the top-ranked Brain Rot countries per development tier (Q13).
- Continue routine, non-urgent monitoring of adult-content exposure, given its currently low prevalence.

Source: `student_brainrot.student_behavior` — analysis performed entirely in SQL (MySQL 8+). Report compiled from documented query logic and findings in the project's analysis script.